

Assignment 2

Mathematics for Cryptography

2025

In this assignment, you practise with small problems (sometimes called toy examples) in number theory and abstract algebra. You should do all (or as much as possible of the) computation by hand unless we request otherwise. This also forces you to discover deeper connections and possible simplifications.

We would like you to explore these problems. The more patterns you discover and interesting observations you make, the more and deeper you learn. That's why it is important that you describe *how* you solve the problems. **Please give explanations not only computations.**

We hope you enjoy this assignment.

Problem 1

- (a) **Prove** that 2 is a generator of the group $(\mathbb{Z}_{19}^*, *)$.
- (b) **Which powers of 2** have order 18?
- (c) After experimenting with a specific cyclic group and its generators, we can prove a general results. **Show** that if a group G has a prime order, then G is cyclic.

Hint: Use Lagrange's theorem.

Problem 2

We will solve the congruence

$$x^{127} \equiv 321 \pmod{557}$$

in a number of steps. All the steps you should carry out manually. Describe your way of thinking.

- (a) **Check** that 557 is prime.
Hint: If 557 is not a prime, its smallest prime divisor is smaller than $\sqrt{557}$.
- (b) **Can we apply** Fermat's little theorem in this case? **What does it tell us** here?
- (c) Describe the group of which the congruence's x is an element.
- (d) To get rid of the exponent of x , we need to find the inverse of 127 modulo $\varphi(557)$. Use the extended Euclidean algorithm to find this inverse.

- (e) Let's call the inverse t , that is, $t \equiv 127^{-1} \pmod{\varphi(557)}$. **Compute** the following exponentiation:

$$(x^{127})^t \equiv ? \pmod{541}.$$

- (f) Why is it true that raising both sides to the power of t , we get the solution of the original congruence?
- (g) **Compute** the solution using the square-and-multiply algorithm. Please include in your description how you performed the modular computations. It's interesting to see your approach, because there are various representations.

Problem 3

- (a) Let $t = \sqrt{5}$ and consider all numbers that can be written as $a + b \cdot t$, where $a, b \in \mathbb{Z}$. For instance, $\sqrt{5}$, 0, or $-3\sqrt{5} - 11$ are such numbers.

To explore these kinds of numbers, first **determine** what a and b are in the aforementioned $-3\sqrt{5} - 11$ and whether the square of this number is also of this form.

To deepen your exploration, **choose** at least three different (a, b) pairs to make new numbers, and **determine their sums and products**. Furthermore, **calculate** the square of each of them. What are the corresponding coefficients (a, b) in the resulting numbers?

Don't forget to **include** how you did the computations. Without understanding the interplay between the operations and these special numbers, it's not possible to truly understand the rest of this problem.

- (b) A more mathematical way to describe the set of numbers in (a) is as follows. $T = \{a + b\sqrt{5} \mid a, b \in \mathbb{Z}\}$. **Show** that $(T, +)$ is a group.

Hint: What are the properties that T has to satisfy under the addition operation to be a group?

- (c) We still consider $T = \{a + b\sqrt{5} \mid a, b \in \mathbb{Z}\}$. **Which group properties** hold for $(T, \cdot)$?

- (d) **Describe** what the answer in (c) tells us about the $(T, +, \cdot)$ being a field?